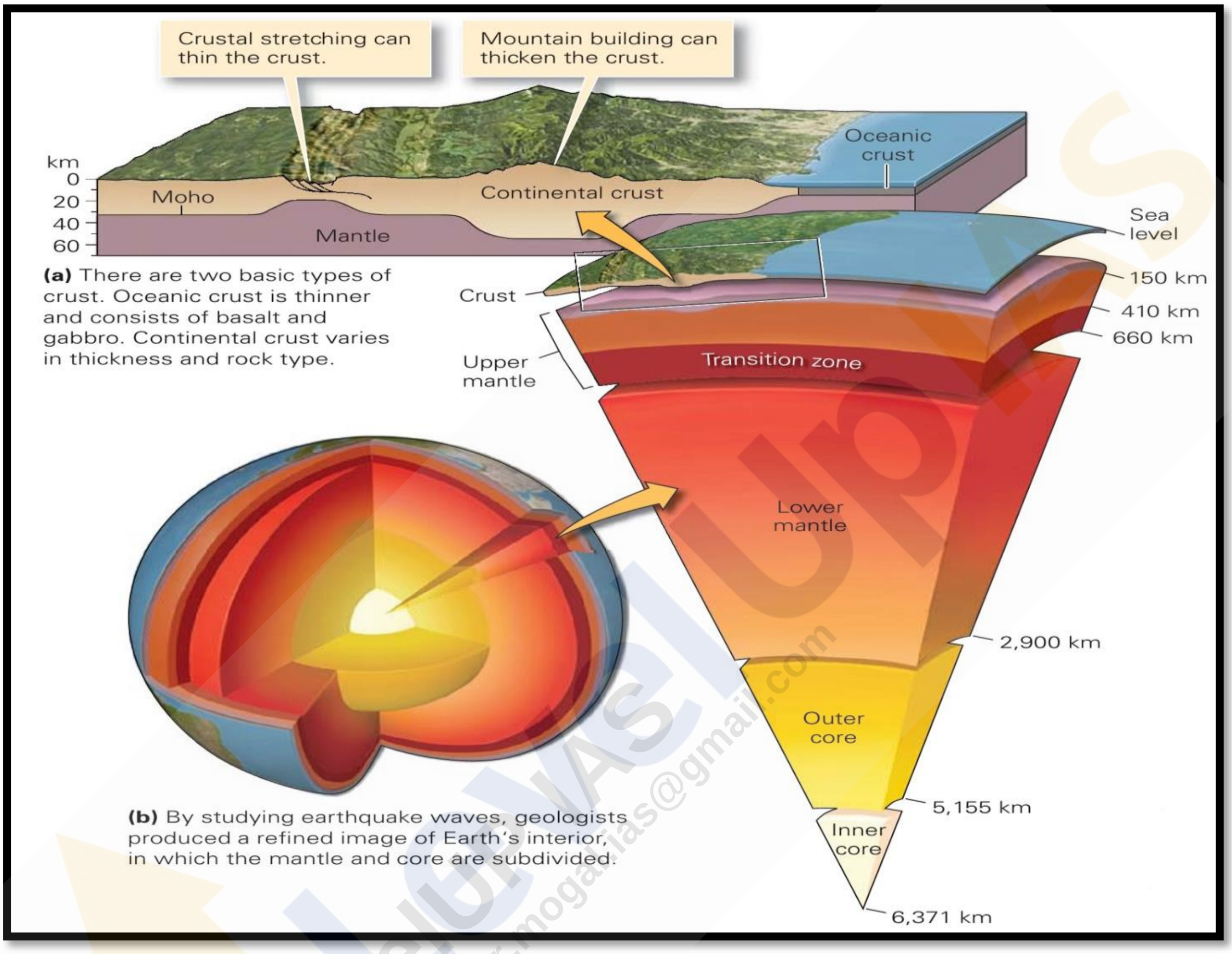
# GEOMORPHOLOGY

## Internal Structure of Earth: Crust, Mantle & Core, Discontinuities

➔ Structure of Earth is divided into four major components:
- Crust
- Mantle
- Outer core
- Inner Core

Each has a unique chemical composition, physical state

Large size, non-uniform structure, high density and temperature make Earth's structure very complex.

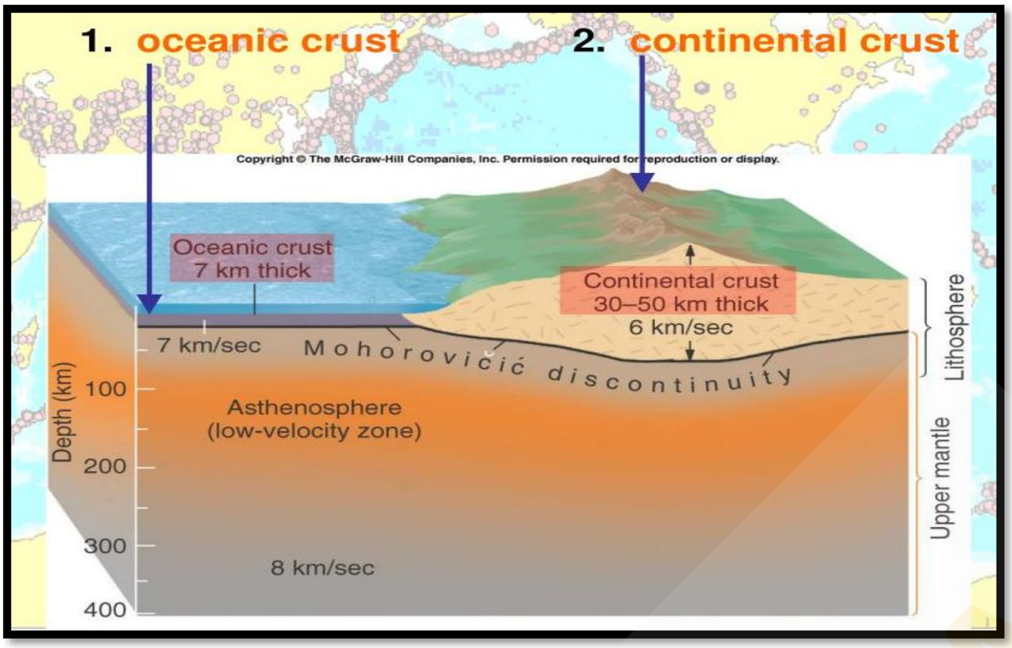



Layers of Earth:-

Crust
- outermost solid part
- Brittle
- OC is thicker than CC

Mantle
- Extends from moho's discontinuity
- Previously divided as upper & lower mantle. But now
- on the basis of discovery by International Union of Geodesy and Geophysics, Mantle is divided into three zones:
  - Zone 1: Moho's discontinuity to 200km depth
  - Zone 2 : 200-700 km
  - Zone 3: 700-2900km
- Velocity of seismic waves relatively slows down in the uppermost zone of upper mantle **"zone of low velocity".**
- Lower mantle extends beyond asthenosphere & is in solid state.

**Lithosphere**: Upper portion of mantel, also called as asthenosphere, which is partially molten



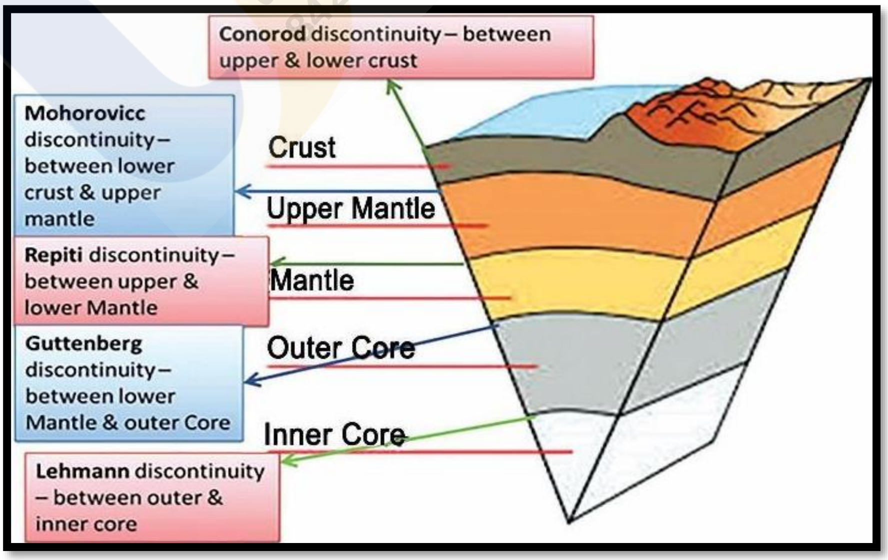

Core
- Outer core is in liquid state while inner core is in solid state.
- S-Waves disappear in outer core.
- Made up of very heavy material (nickel and iron) – nife layer.
- Core is responsible for generation of Earth's magnetic field.
- Contains information regarding earliest history of accretion of planet.

<u>Discontinuity within earth</u>

- Unique layers within the Earth are separated from each other through a transition zone. These transition zones are called **discontinuities.**
- They are marked by variations in density of material, velocity of seismic waves, temperature and pressure conditions.
- There are five discontinuities inside the earth:

(i) **Conrad Discontinuity:** Between SIAL and SIMA.
(ii) **Mohorovic Discontinuity:** Between the Crust and Mantle.
(iii) **Repiti Discontinuity:** Between Outer mantle and Inner mantle.
(iv) **Gutenberg Discontinuity:** Between Mantle and Core.
(v) **Lehman Discontinuity:** Between Outer core and Inner core.

**C M R G L**



## Sources of Information About Earth's Interior

- Direct
  - mining
  - volcanic eruption
- Indirect
  - Change in pressure & temperature
  - Seismic waves
  - Meteorites
  - Gravitation
  - Magnetic field

Meteoroids
Meteors
meteorites
Explained in class

## Theories explaining distribution of Oceans and Continents

1. Continental Drift Theory

- Alfred Wegner in 1912
- Also called as 'displacement hypothesis'
- Believed in three-layer system – outer, intermediate, lower.
- All continents formed a single continental mass (PANGEA) & mega ocean (PANTHALASSA) surrounded the same.
- Pangaea first broke into two large continental masses as Laurasia & Gondwanaland.
- Laurasia and Gondwanaland continued to break into various smaller continents that exist today.
- Force for Drifting is responsible for the drifting of the continents, was caused by pole-fleeing force and tidal force
  - pole-fleeing force: Related to rotation of the earth.
  - tidal force: Force due to attraction of moon

- Evidence
  - Rocks of Same Age Across the Oceans
  - Gondwana system of sediments from India is known to have its counterparts
  - Placer Deposits
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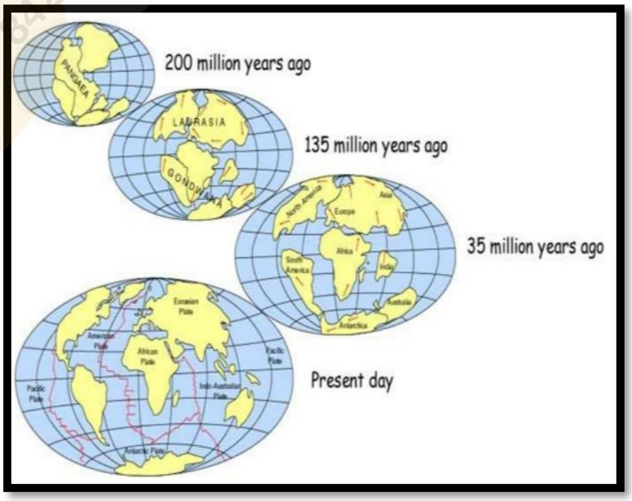
  - Carboniferous glaciation
  - Seafloor spreading & Plate tectonic theory

Paleomagnetic studies
Seafloor spreading
Polar wandering
Explained in class

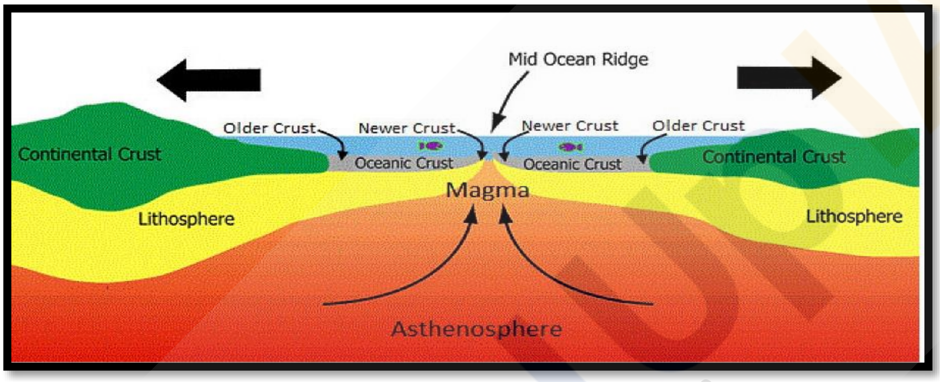


2. Sea floor spreading theory

- Prof Harry Hess.
- Mid oceanic ridges are situated on the rising thermal convection currents coming up from the mantle.
- Constant eruptions cause the rupture of the oceanic crust and the new lava wedges into it, pushing the oceanic crust on either side. **The ocean floor, thus spreads.**
- Molten lava cools down and solidifies
- Spreading of one ocean does not cause the shrinking of the other because of the consumption of the Oceanic crust at the oceanic trenches.

Evidence

- **Molten material**
- **Seafloor drill**
- **Radiometric age dating and fossil ages**
- **Magnetic stripes**
- **Very thin sediments on ocean floor**

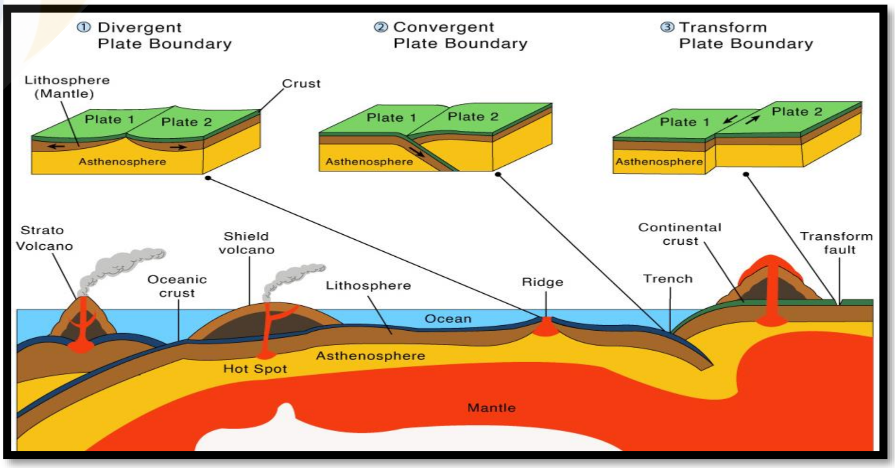



3. **Plate tectonic theory**

- by McKenzie and Parker in 1967 and outlined by Morgan in 1968.
- Tectonic plate is a massive, irregularly shaped slab of solid rock,
- composed of both continental and oceanic lithosphere.
- Plates move horizontally
- Forces responsible-- heat beneath the earth is generated because of (i) Radioactive decay (ii) Residual heat.
- Three types pf plate boundaries-- (i) Divergent (ii) Convergent (iii) Transform

Evidence

- Distribution of fossils
- Occurrence of Earthquakes
- Location of continental and ocean floor features



# Plate Tectonics & Earth Surface: Movement of Indian plate, East African Rift Valley & Earthquakes in India

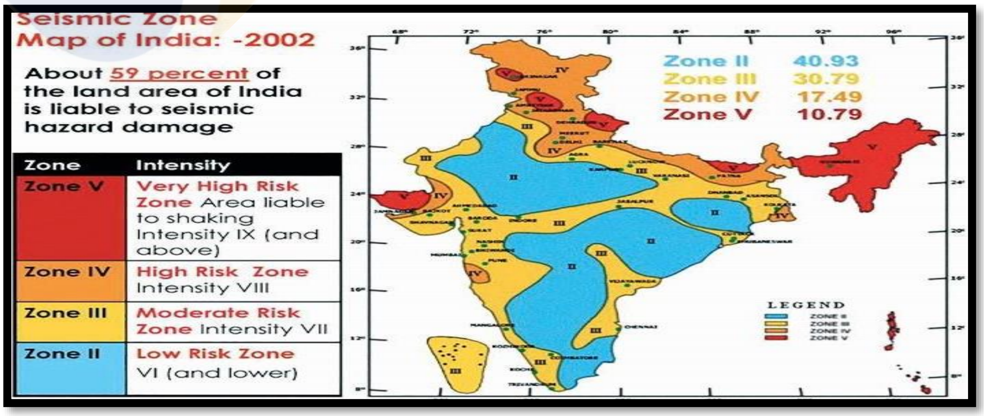
1. Movement of Indian Plate:
   - Subduction zone along Himalayas forms northern plate boundary in the form of continent—continent convergence.
   - East, it extends through Arakan yoma Mountains
   - Western margin follows Kirthar Mountain, Western margin follows Kirthar Mountain  and joins the Red Sea rift south eastward along the Chagos Archipelago.
   - Boundary between Indo-Australian and Antarctic plate is marked by oceanic ridge (divergent boundary)
   - The Tethys Sea separated India from the Eursian continent.
   - India started her northward journey & at the time when Pangaea broke. Tibetan block was closer to the Asiatic landmass.
   - India collided with Eurasia about 40-50 Mn years ago causing rapid uplift of the Himalayas (Still rising).

2. Plate tectonics and East African rift valley:
   - East-African Rift Valley (EAR) is a developing divergent plate boundary
   - Eastern portion of Africa, Somalian plate, is pulling away from rest of continent, which comprises Nubian plate
   - Numalian plates are also separating from Arabian plate in thus creating a ‘Y’ shaped rifting system.

3. Plate tectonics & earthquakes:
   - Major tectonic events associated with plate boundaries are ruptures and faults, faulting and folding & transform faults
   - At divergent boundaries, moderate earthquakes occur because the rate of rupture of the crust and upwelling of lavas due to fissure is also slow.
   - At convergent boundaries, earthquakes of high magnitude and deep foci occur because of deeper subduction of one plate. mountain building, faulting and violent volcanic eruptions cause disastrous earthquake.

Distribution of Earthquakes

- World
  - Circum-pacific belt
  - Mid-Atlantic belt
  - Mid- continental belt
- India
  - Divided into zones very high risk, high risk, moderate risk, low risk



Reasons for high seismicity in Himalayas

- Active C-C convergence zone
- Double layering' effect produces shallow foci earthquakes
- Region of young, folded mountains with sedimentary rock formations have weaker zones and numerous faults.
- Stress been accumulated for over thousands of years
- Steep slopes lead to frequent landslides in turn leading to earthquakes.
- Anthropogenic activities.

**Earthquake & Shadow zones**

Earthquake waves or seismic waves are vibrations generated by an earthquake, explosion, or similar energetic source and propagated within Earth or along its surface.

Types of waves: (i) Body waves (ii) Surface waves

(i) Body waves

- Travel in all directions within the body
- Velocity as well as direction of waves changes as they travel through materials.
- Two types : P & S waves.

| **P- waves (primary waves)** | **S-waves (secondary waves)** |
|---|---|
| Move faster and arrives first at the surface. | Arrive at the surface with some time lag |
| Travels through all mediums. | Travels only through solid medium |
| Vibrates Similar to longitudinal waves. (parallel to direction of wave) | Vibrates Similar to transverse waves. (perpendicular to direction of wave) |
| Also called as 'compressional waves' | Also called as 'distortional waves' |

(ii) Surface Waves

- Body waves when interact with the surface rocks, they generate new waves called as surface waves.
- Also called as long period waves.
- Last to report on seismograph, but they cover longest distance.
- More destructive.

Concept of Shadow zones

**Shadow zone**: Areas where the seismographs do not record any earthquake waves.

< 105 degrees = No shadow zone for P as well as S waves (Both the waves are recorded by seismograph)
105-145 degree = Shadow zones for both (Both P & S waves are not recorded)
> 145 degrees = Shadow zone for S waves (P waves recorded)

## How Seismic waves help to know about Earth's interior?

(i) Non-linear travel path → Earth's structure in not homogeneous

(ii) Curved path → shows increase in density as one moves deeper

(iii) change of velocity → help us to infer that there are three zones/layers of varying densities inside the Earth

(iv) Bouncing back of waves → help to determine the depth of the discontinuities and thickness of various layers.

(v) Absence of S-waves in the core → confirms the presence of liquid core

(vi) velocity decreases in upper part of the upper mantle → confirms the presence of lighter material in between.

## **Forces acting on Earth**

- Movements of earth
  - Endogenic
    - Diastrophism
      - Epeirogenic
        - Upward
        - Downward
      - Orogenic
        - Tension
        - Compression
    - Sudden movements
      - Earthquakes
      - Volcanoes
  - Exogenic
    - Weathering
      - Physical
      - Chemical
      - Biological
    - Erosion
      - Ice
      - River water
      - Sea
      - Wind
      - Underground water

### Endogenic Forces:

- Land building forces.
- lead to vertical and horizontal movements →result in subsidence, land upliftment, volcanism, faulting, folding, earthquakes, etc.
- These forces are created by Primordial heat, radioactivity, tidal and rotational friction from the earth.
- Epeirogenic → vertical forces, responsible for mountain building, can lead to submergence as well as divergence.
- Orogenic → horizontal forces, responsible for mountain building
- Catastrophic forces/ sudden movements → fast movements.

### Exogenic forces:

- Land modifying forces.
- Cause land to wear down → land wearing forces/denudational forces.
- Performed by geomorphic agents (wind, water, waves etc).
- Source of power → solar energy.

## Volcanism

- volcano is a vent or fissure in the crust through which lava, ash, rock and gases erupt.
- Sometimes, steam and gas emit before the emission.
- Volcano erupts→Magma moves upward→vent or conduit→flows out from crater.
- Erupted material forms as a volcanic mountain.

→Consist of layers of solidified lava (lava flows)

→Includes fragments thrown from the volcano.

Difference between lava and magma → → → →

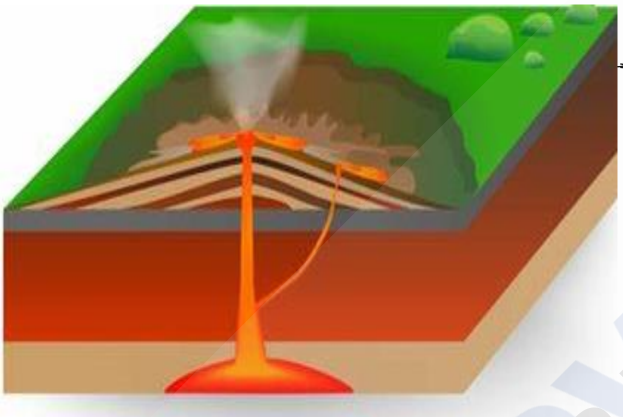
Will be explained in class

Causes of volcanism:

→The buoyancy of the magma.
→Pressure from the exsolved gases in the magma.
→Increase in pressure on the chamber lid.

Types of Volcanoes:

(i) Shield volcanoes
(ii) Cinder cone volcanoes
(iii) Composite volcanoes
(iv) Caldera.



Shield volcano

→Not very steep but far and wider.
→ Extend to great height and distance.
→ Largest of all volcanoes ← lava flows far distance.
→ Low slopes, mostly frozen lava.
→ Made up of basalt → less viscous
→ Low explosive (turns explosive when water gets into vent.
→ Develops cinder cone.

Cinder Cone Volcanoes

→ Cinder – extrusive igneous rocks
→ Small volcanoes.
→ Contains loose, grainy cinder and almost no lava.
→ Very steep sides, small craters on top.
→ Moderate steep sides, small craters.

Composite Volcanoes

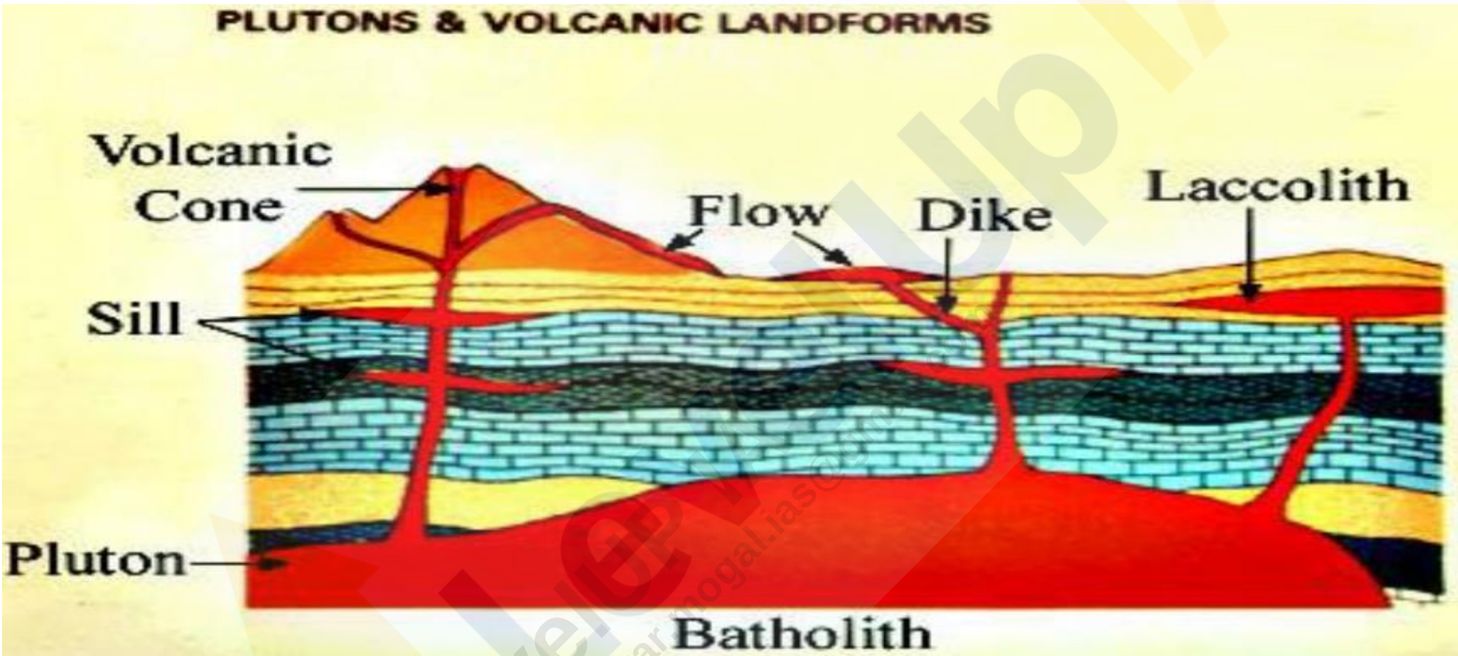
→ Called as strato volcano → it consists of layers of
→ solid lava mixed with sand or gravel.
→ Erupts cooler and viscous lava.
→ Explosive eruptions.


- Most explosive.
- Collapse themselves and collapsed depressions are called as calderas.
- Large magma chamber, close vicinity → explosive nature.

<u>Volcanic Landforms:</u> (i) Extrusive (ii) Intrusive

(i) Extrusive
- called as volcanic Igneous rocks
- Cools at surface
- Ex. Basalt, Andesite.

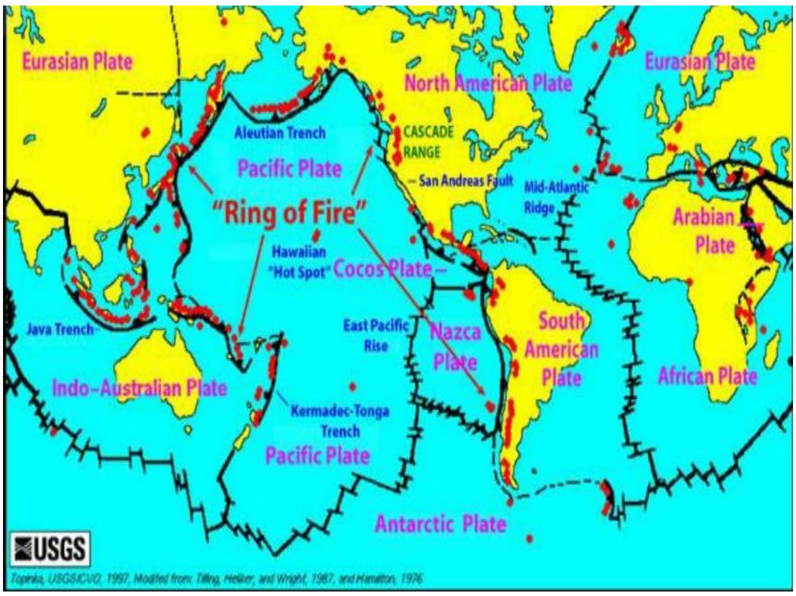
(ii) Intrusive
- also called as plutonic igneous rocks
- Cools in the crust.
- Ex. Granite, gabbro, diorite




<u>Distribution of volcanoes:</u>

- Circum-pacific belt.
- Mid-world mountain belt.
- African rift valley belt.

> Volcanoes—folding & faulting region.
>
> Occur along coastal mountain ranges, island and mid oceans
>
> Most of the active volcanoes found—pacific region (pacific ring of fire)



Hot water springs in India

- Also called as thermal springs.
- Formed by the geothermally heated water emerging through cracks.
- Heated water – less dense → rise upwards.
- Useful for both recreational and therapeutic.

Ex. Manikaran , HP, Vashisht, HP, Surya kund, Uttarakahand, Gaurikund, Uttarakhand Tataoani, Chhattishgarh, Unnao, UP.

Why there are no volcanoes in Himalayas?

(a) Less rise in temperature as subduction angle is not much ← C-C type convergence.
(b) No oceanic plate → not heated water to form vapours.
(c) Increase in crustal thickness due to double layering → not allow magma to outpour.

Effect of volcanism on climate:

Cooling effect

Warming effect

Ozone layer depletion → → → → → How?

Will be explained in the class.